



22AIE301 Probabilistic Reasoning

Customer Behavior Analysis & Simulation with Markov Models

Group C-14

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CERTIFICATE

This is to certify that we, the student of Amrita School of Artificial Intelligence, has completed the “*Customer Behavior Analysis & Simulation with Markov Models*” as part of their *Cloud Computing* Course Project.

The project work was carried out under the guidance and supervision of Dr. Milton Mondal, Assistant Professor, Amrita School of Artificial Intelligence, Coimbatore. To the best of our knowledge this work has not formed the basis for the aware of any degree/diploma/ associate ship/fellowship/ or a similar award to any candidate in any University.

Date:

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Acknowledgement

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Abstract

This project presents a data-driven simulation of customer movement within a supermarket using a probabilistic model derived from real behavioral data. The simulation employs a Markov Chain-based transition matrix created from customer location logs collected over multiple weekdays. Each customer's path is modeled as a stochastic process, determining transitions between store sections such as drinks, dairy, spices, fruit, and checkout. Using OpenCV, a graphical simulation visualizes real-time customer movement, section visits, and revenue generation. The model helps identify which sections attract the most customers and contribute the highest revenue, providing insights for optimal store layout design. By combining data analytics and visual simulation, this project demonstrates how customer flow patterns can be quantified and improved for enhanced store efficiency and profitability.

Problem Statement

- *Retailers often face challenges in understanding how store layout affects customer movement and sales. Without proper data-driven insights, product placement and section arrangements may fail to optimize revenue or shopping convenience.*
- *The problem addressed in this project is to analyze and simulate customer flow within a supermarket using actual movement data, enabling identification of high-traffic and high-revenue sections.*
- *By applying a probabilistic model, the project aims to suggest an optimal layout configuration that enhances customer experience and improves overall store efficiency.*

Introduction

Understanding customer behavior within retail environments is crucial for improving store design, optimizing product placement, and maximizing sales. Traditional methods for studying shopping patterns rely on manual observation or static analytics, which often fail to capture the dynamic nature of in-store movements. This project introduces a simulation framework that uses real-world customer movement data to model shopping behavior probabilistically.

Data collected from customer location logs over five weekdays form the foundation of the analysis. Using these logs, a transition matrix is computed to capture the probability of moving from one store section to another. This matrix forms the basis of a Markov Chain model that drives the movement of virtual customers in a simulated supermarket environment. The simulation, developed in Python using OpenCV, visualizes live customer traffic, section visits, and generated revenue. The approach provides valuable insights into how layout arrangements influence customer flow and profitability, assisting retailers in data-backed decision-making.

Mathematics Involve

A Markov chain is a way to describe a system that moves between different situations called "states", where the chain assumes the probability of being in a particular state at the next step depends solely on the current state. This key property of ignoring historical states is called the Markov Property or memory lessness.

Fundamental Properties

1. Irreducibility: *It is possible to reach any state from any other (maybe through several steps).*

2. Periodicity: *A state is periodic if returns are only possible at multiples of some step (>1).*

3. Transience and Recurrence:

- *Transient: The system might never return to this state.*
- *Recurrent: The system is certain to return eventually.*

4. Absorbing States: *Once entered, the process cannot leave i.e. no outgoing transitions.*

Discrete-time Markov chain

Main article: [Discrete-time Markov chain](#)

A discrete-time Markov chain is a sequence of [random variables](#) $X_1, X_2, X_3, \dots$ with the [Markov property](#), namely that the probability of moving to the next state depends only on the present state and not on the previous states:

$$\Pr(X_{n+1} = x \mid X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \Pr(X_{n+1} = x \mid X_n = x_n), \quad \text{if}$$

both [conditional probabilities](#) are well defined, that is, if $\Pr(X_1 = x_1, \dots, X_n = x_n) > 0$.

The possible values of X_i form a [countable set](#) S called the state space of the chain.

Variations

- *Time-homogeneous Markov chains are processes where*

$$Pr(X_{n+1} = x \mid X_n = y) = Pr(X_n = x \mid X_{n-1} = y)$$

for all n. The probability of the transition is independent of n.

- *Stationary Markov chains are processes where*

$$Pr(X_0 = x_0, X_1 = x_1, \dots, X_k = x_k) = Pr(X_n = x_0, X_{n+1} = x_1, \dots, X_{n+k} = x_k)$$

for all n and k. Every stationary chain can be proved to be time-homogeneous by Bayes' rule.

A necessary and sufficient condition for a time-homogeneous Markov chain to be stationary is that the distribution of X_0 is a stationary distribution of the Markov chain.

- *A Markov chain with memory (or a Markov chain of order m) where m is finite, is a process satisfying*

$$\begin{aligned} & Pr(X_n = x_n \mid X_{n-1} = x_{n-1}, X_{n-2} = x_{n-2}, \dots, X_1 = x_1) \\ = & Pr(X_n = x_n \mid X_{n-1} = x_{n-1}, X_{n-2} = x_{n-2}, \dots, X_{n-m} = x_{n-m}) \text{ for } n > m \end{aligned}$$

In other words, the future state depends on the past m states. It is possible to construct a chain (Y_n) from (X_n) which has the 'classical' Markov property by taking as state space the ordered m-tuples of X values, i.e., $Y_n = (X_n, X_{n-1}, \dots, X_{n-m+1})$.

Methodology

1. Data Collection and Preprocessing

The initial data was collected over five weekdays (Monday to Friday) in CSV format, containing the following fields: timestamp, customer_no, and location.

Each record indicates the position of a customer in a particular store section at a specific time. These datasets were merged into a single unified file to represent overall customer activity across the week. The data was then sorted chronologically by timestamp and grouped by customer_no to track the sequential movement of each customer through the supermarket.

*From this processed data, a new dataset named **path.csv** was created, where each customer's path (e.g., entry → drinks → dairy → checkout → exit) represented their shopping sequence. This step ensures that the temporal and spatial order of movements is preserved for subsequent modeling.*

2. Construction of Transition Matrix

*To model the probabilistic movement of customers between different store sections, a **Markov Chain-based Transition Matrix** was constructed from the path.csv file.*

*Each unique section in the store—entry, drinks, dairy, spices, fruit, checkout, exit—was treated as a **state**. For every customer path, consecutive transitions (e.g., entry → drinks, drinks → dairy) were counted. The frequency of each transition was then converted into a probability by normalizing over the total outgoing transitions from each state.*

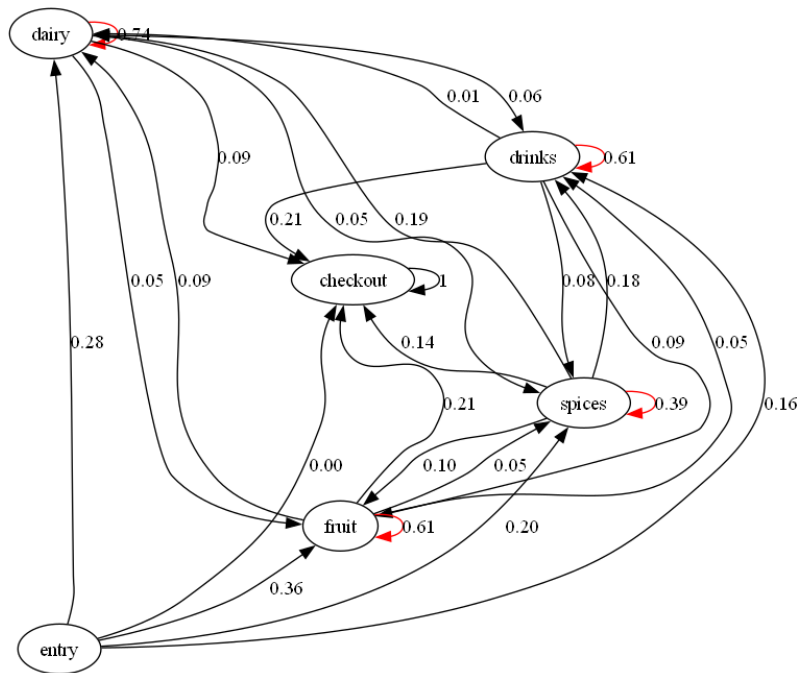


Fig 1: The image is a diagram of a Markov chain generated from a transition matrix in the file *transition.csv*, which shows probabilities of moving between states, such as sections in a supermarket.

3. Simulation Design

The simulation was implemented in **Python** using **OpenCV** for real-time visualization. It consists of two main classes:

- Customer** **Class:**
Represents individual customers. Each customer is assigned an initial position near the store entry and a random target section determined by the probabilities in the transition matrix. The customer moves frame-by-frame toward their target section, waits for a short period (simulating browsing time), and then probabilistically chooses the next section based on the transition matrix. Once a customer reaches the checkout section and completes the counter duration, they exit the simulation.
- Supermarket** **Class:**
Represents the store environment, including the floor layout, counters, and section boundaries. It also tracks section-wise visits, elapsed time, and revenue

generation. The `draw()` function updates the display with customer positions, section visit counts, and total revenue.

4. Visualization and Revenue Tracking

During simulation, the following key parameters are displayed on the OpenCV window:

- **Elapsed Time:** Time since simulation start.
- **Section Visit Counts:** Real-time count of how many times customers have entered each section.
- **Revenue:** Computed by multiplying the number of visits to each section with predefined section-wise revenue values (e.g., drinks = \$5, dairy = \$4, spices = \$2, fruit = \$3).

5. Result Interpretation and Layout Recommendation

When all customers exit the store or when the user stops the simulation, a detailed summary is printed in the console, including:

- Total visits per section
- Individual and total revenue per section
- Identification of the section generating the highest revenue (excluding checkout)

Based on this information, the simulation suggests **which section should be relocated closer to the entry** to optimize customer engagement and revenue potential.

Results

The simulation successfully modeled and visualized customer movement within the supermarket based on real transition probabilities derived from the dataset. Each virtual customer followed a distinct probabilistic path through store sections such as **drinks**, **dairy**, **spices**, **fruit**, and **checkout**, reflecting realistic shopping behavior patterns.

	Entry	Dairy	Spices	Drinks	Fruit	Checkout
Entry	0	0.287441236	0.181329752	0.153391538	0.377300201	0.000537273
Dairy	0	0.737022665	0.051346943	0.058545639	0.049772229	0.103312525
Spices	0	0.193214399	0.402198152	0.16310927	0.090952533	0.150525645
Drinks	0	0.010899743	0.086992288	0.598560411	0.087917738	0.21562982
Fruit	0	0.095743844	0.050664779	0.054834395	0.597199276	0.201557706
Checkout	0	0	0	0	0	1

Table - The Markov transition matrix provided in a table format, representing the probabilities of moving between states (entry, dairy, spices, drinks, fruit, checkout)

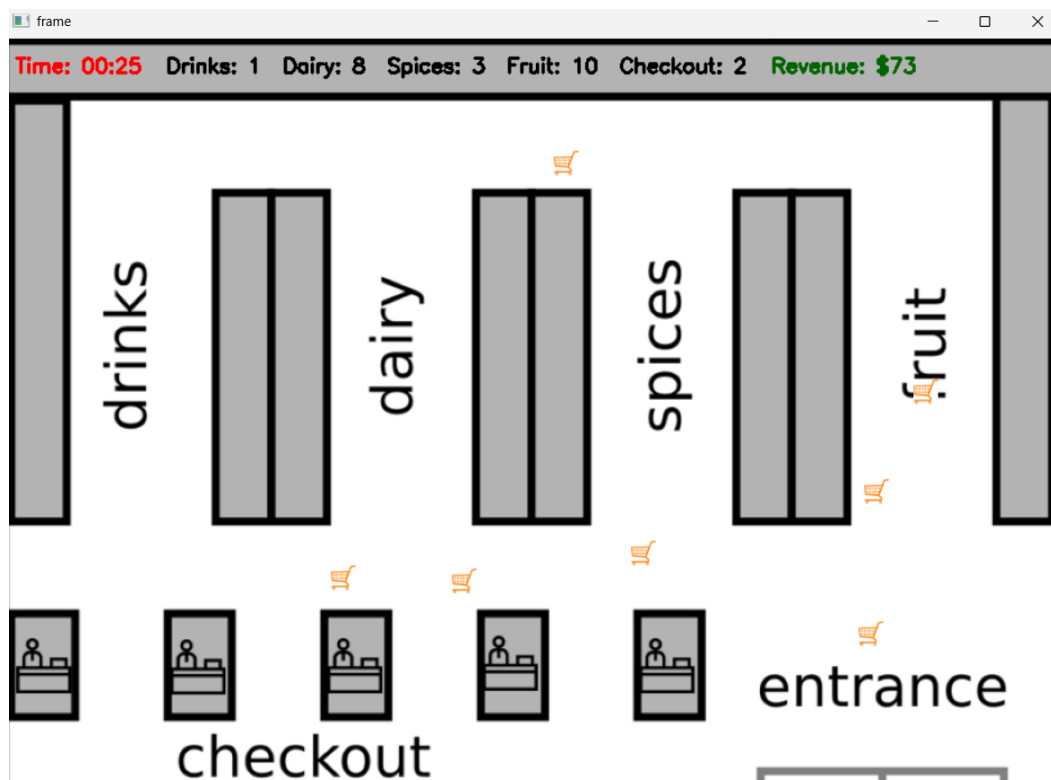


Fig 2 - Simulates customer movement in a supermarket using a Markov transition matrix,

During execution, the OpenCV visualization dynamically displayed the **elapsed simulation time, number of visits per section, and real-time revenue generation**. Customers were observed entering the store, moving toward selected sections, pausing briefly to simulate browsing, and finally proceeding to checkout before exiting. The system-maintained section-wise visit counters, updated revenue values, and generated a continuous visual representation of customer flow.

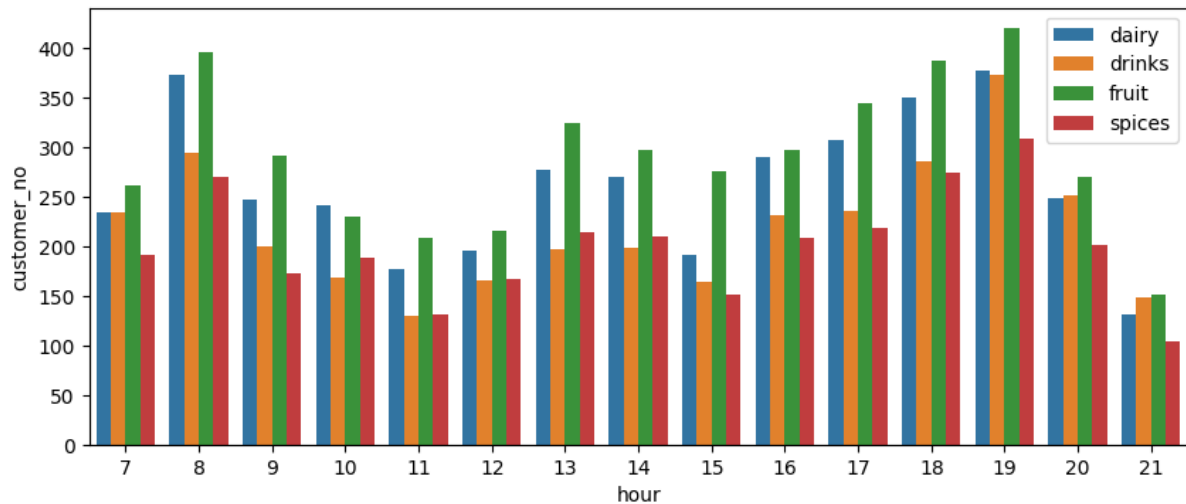


Fig 3 - This Histogram tells the average number of people on each section on a weekly data

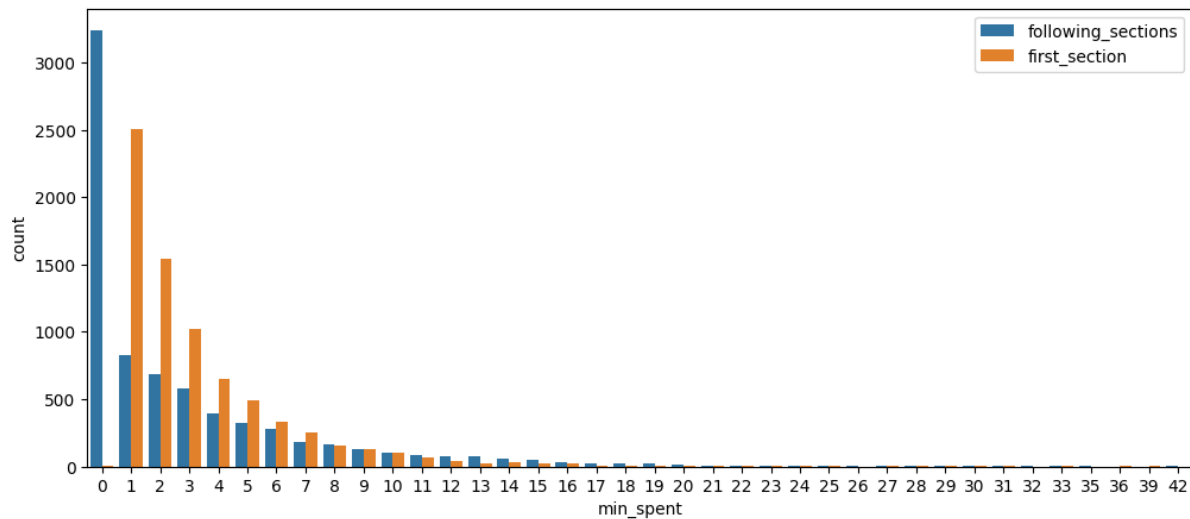


Fig 4 - This plot tells number of customers visiting the first section compared with following sections so that we can set first section efficiently.

At the end of the simulation, a detailed statistical summary was produced in the console, reporting:

- *The **number of visits** recorded for each section*
- *The **revenue contribution** of each section based on predefined values*
- *The **total revenue** generated during the simulation period*
- *The **highest revenue-generating section**, which was recommended for relocation closer to the store entry to increase accessibility and maximize profits*

*The results demonstrated that sections with higher visit frequencies, such as **drinks** or **fruit**, contributed significantly to total revenue. This validated the effectiveness of the transition-matrix-based approach for predicting and optimizing customer movement patterns. The simulation thus provides actionable insights for **store layout optimization** and **revenue enhancement** using a data-driven methodology.*

Conclusion

*This project successfully demonstrated how customer behavior within a supermarket can be modeled and visualized using a data-driven, probabilistic approach. By deriving a transition matrix from real customer movement data, the simulation accurately replicated realistic shopping patterns through a Markov Chain framework. The OpenCV-based visualization provided clear insights into section-wise visits, time spent, and revenue contribution. The analysis revealed that sections with higher transition probabilities from the entry point generated greater revenue, emphasizing the importance of strategic section placement. Overall, the project highlights the potential of combining **data analytics, probability modeling, and computer vision** to optimize supermarket layouts and enhance decision-making for retail management.*

References

- [1] J. R. Norris, Markov Chains. Cambridge, U.K.: Cambridge University Press, 1997.*
- [2] S. M. Ross, Introduction to Probability Models, 11th ed. Amsterdam, Netherlands: Academic Press, 2014.*
- [3] A. M. Law, Simulation Modeling and Analysis, 5th ed. New York, NY, USA: McGraw-Hill Education, 2015.*
- [4] C. M. Bishop, Pattern Recognition and Machine Learning. New York, NY, USA: Springer, 2006.*
- [5] W. Feller, An Introduction to Probability Theory and Its Applications, vol. 1. Hoboken, NJ, USA: Wiley, 2008.*
- [6] OpenCV Team, “OpenCV Documentation – Open Source Computer Vision Library,” 2024. [Online]. Available: <https://docs.opencv.org>*
- [7] Pandas Development Team, “Pandas: Powerful Python Data Analysis Toolkit,” 2024. [Online]. Available: <https://pandas.pydata.org>*
- [8] NumPy Developers, “NumPy: Fundamental Package for Scientific Computing in Python,” 2024. [Online]. Available: <https://numpy.org>*
- [9] J. Banks, J. S. Carson, B. L. Nelson, and D. M. Nicol, Discrete-Event System Simulation, 5th ed. Upper Saddle River, NJ, USA: Prentice Hall, 2010.*
- [10] M. E. Lantz, Machine Learning with Python for Everyone. Hoboken, NJ, USA: Pearson Education, 2019.*